Valid types of switch expression:

---

byte
short
int
char
Byte
Short
Integer
Character
enum (JDK 1.5)
String (JDK 1.7)

Invalid types are:

---

long
float
double
boolean

Some important points:

---

a) We cannot accept **duplicate** case value.

b) The label of case must be **constant OR literals OR expression**.

c) The break statement is used to exit from the switch block. It is optional but recommended [JDK 17 (->) break is not required]

d) The default case is **optional** and executes if no case matches in the switch expression as well as we can write default case anywhere in the switch.

//Programs on switch case :

SwitchDemo.java [Old Style]

```
//switch case

void main()
{
     int choice = Integer.parseInt(IO.readln("Enter your choice :"));

     switch(choice)
     {
          case 1 :
               IO.println("Case 1");
          break;
          case 2 :
               IO.println("Case 2");
          break;
          case 3 :
               IO.println("Case 3");
          break;
          default : IO.println("Invalid Case");
     }
}
```

SwitchDemo1.java

```
/* Write a switch case program where based on the first character
select the color */

void main()
{
   char ch = IO.readln("Enter Color Name :").toUpperCase().charAt(0);

   switch(ch)
    {
        case 'R' -> IO.println("RED Color");
        case 'W' -> IO.println("WHITE Color");
        case 'B' -> IO.println("BLUE Color");
        case 'G' -> IO.println("GREEN Color");
        default -> IO.println("Invalid color choice");
    }

    IO.println("Out of switch body");
}
```

SwitchDemo2.java

```
void main()
 {

        IO.println("\t\t**Main Menu**\n");
          IO.println("\t\t**100 Police**\n");
          IO.println("\t\t**101 Fire**\n");
          IO.println("\t\t**102 Ambulance**\n");
          IO.println("\t\t**139 Railway**\n");
          IO.println("\t\t**181 Women's Helpline**\n");

          IO.print("Enter your choice :");
          int choice = Integer.parseInt(IO.readln());

          switch(choice)
          {
          case 100 ->  IO.println("Police Services");

          case 101 ->  IO.println("Fire Services");

          case 102 -> IO.println("Ambulance Services");

          case 139 -> IO.println("Railway Enquiry");

          case 181 -> IO.println("Women's Helpline ");

          default ->     IO.println("Your choice is wrong");

      }
 }
```

The same program we can re-write with the help of an infinte loop to display the menu.

```
 void main()
 {
    while(true)
      {
          IO.println("\t\t**Main Menu**\n");
          IO.println("\t\t**100 Police**\n");
          IO.println("\t\t**101 Fire**\n");
          IO.println("\t\t**102 Ambulance**\n");
          IO.println("\t\t**139 Railway**\n");
          IO.println("\t\t**181 Women's Helpline**\n");

          IO.print("Enter your choice :");
          int choice = Integer.parseInt(IO.readln());

          switch(choice)
          {
          case 100 ->  IO.println("Police Services");

          case 101 ->  IO.println("Fire Services");

          case 102 -> IO.println("Ambulance Services");

          case 139 -> IO.println("Railway Enquiry");

          case 181 -> IO.println("Women's Helpline ");

          default ->     IO.println("Your choice is wrong");

      }

     String userChoice = IO.readln("Do u want to continue [yes/no] :");
       if(userChoice.equalsIgnoreCase("no"))
           break;


      }

 }
```

SwitchDemo3.java

```
 //Passing String in the switch case [Allowed from JDK 1.7]

 void main()
 {
      String season = IO.readln("Enter the name of the season :").toLowerCase();

      switch(season)  //JDK 1.7
      {
           case "summer" -> IO.println("Summer season");
           case "rainy" -> IO.println("Rainy season");
           case "winter" -> IO.println("Winter season");
           case "spring" -> IO.println("spring season");
           default -> IO.println("Invalid season");
      }

 }
```

Switch Case with Grouping in Java:

In switch statement, It is possible to combine **multiple case labels** to execute one common block of code. It must be separated by , (comma)

We should use when **multiple values (cases)** should result in the same action.

Grouping.java

```
//switch with grouping

void main()
{
     int choice = Integer.parseInt(IO.readln("Enter your choice 1 - 10 :"));

     switch(choice)
     {
          case 1,3,5,7,9 -> IO.println("Odd Number");
          case 2,4,6,8,10 -> IO.println("Even Number");
     }
}
```

SwitchDemo4.java

```
//Grouping
void main()
{
     IO.print("Enter an animal Name :");
     String animal = IO.readln().toLowerCase();

   switch(animal)
     {
          case "cat", "dog", "cow" ->
          {
               IO.println("Domestic Animal");
          }

          case "lion", "tiger", "fox" ->
          {
         IO.println("Wild Animal");
          }

     }

}
```

Switch Case with Return Value (Expression Style):

From JDK 14V onwards, A switch statement can also **return some value** that means we can use switch statement as **switch expression**.

This is known as switch case with return value OR expression style switch.

Rules:

---

a) While writing switch body ; (semicolon is compulsory at the end)
b) Here default -> is compulsory [If no case will match then default will return value]
c) From the switch body, we need to return **appropriate value based on type**.
d) If, we have multiple statements ({} using block) then
   yield keyword is compulsory to return the value explicitly
   [In a single statement, value will return automatically]

SwitchDemo5.java

```
//Based on the student grade return the student result

void main()
{
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' -> "Excellent!!!";
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```

Case 1 :

```
//Based on the student grade return the student result

void main()
{
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' -> "Excellent!!!";
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      }  //Semicolon is compulsory here

      IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```

Case 2 :

```
void main()
{
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' -> "Excellent!!!";
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";  //Writing defult is compulsory
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```

Case 3 :

```
void main()
{
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)      Automatically will assign to result which is of String type
      {                                for a SINGLE STATEMENT
      case 'A', 'B' -> "Excellent!!!";
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```

Case 4 :

```
//Based on the student grade return the student result

void main()
{
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' ->
          {
          yield "Excellent!!!";  //{} reprsents multiple statements so yield keyword is compulsory to return the
                                                                                            value expliciltly
          }
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```

Case 5:

```
//Based on the student grade return the student result

void main()
{
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' -> yield "Excellent!!!";  //for single statement yield is not reqd
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```

Case 5

```
//Based on the student grade return the student result

void main()
{
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' ->
          {
            yield "Excellent!!!";
            yield "Super!!!";  //Unreachable only one yield is allowed
          }
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```